

PCR-Net: Command-Space Conflict Arbitration for Hexapod Target Following in Clutter

Abstract—Target following in clutter is not merely tracking a moving waypoint: the target’s traversable space can differ from the robot’s. For a legged robot, this passability mismatch turns local following into a command-space arbitration problem, because the command that preserves following can be incompatible with the command that keeps the robot collision-free. PCR-Net formulates this conflict as continuous arbitration within a single executable velocity command, rather than a switch between separate behaviors. It decomposes the command into Follow and Avoid proposals and fuses them through a risk-conditioned learned gate on top of a fixed low-level locomotion policy. On a controlled target-following benchmark in structured clutter, PCR-Net sustains 0.95 task success at an above-training target speed, while rule-based, structural, learning-based, and planner-style alternatives expose distinct safety-following failure modes. A checkpoint-controlled inference ablation shows that the learned correction supports target retention during high-conflict maneuvers. Across 60 real-robot trials, PCR-Net succeeds in 37/40 runs, compared with 12/20 for Rule-Override, using onboard target detection and stereo-depth maps in corridor clutter.

I. INTRODUCTION

A legged robot following a moving person faces a difficulty that is especially acute for a finite-bodied platform with gait dynamics: *the target’s passability is not the robot’s safe passability* (Fig. 1). A person can turn sideways through a narrow gap or step between obstacles, whereas the robot’s wider gait envelope cannot safely reproduce the same path. The path that keeps the robot closest to the person may lead it into an obstacle, while the path that keeps it safe may cause it to lose the person. The task is therefore neither pure obstacle avoidance nor pure target tracking.

We study this difficulty as a conflict in the robot’s *command space*. At each control step the high-level controller emits one velocity command. The command that best maintains following and the command that best avoids collision can become incompatible—competing for command authority within that single output. Standard local navigation formulations that treat the target only as a moving waypoint do not explicitly preserve target tracking during local avoidance, and fixed-priority switching between a follow and an avoid behavior can lose the target during the maneuver. We therefore study an arbitration that reconciles the two objectives within a single command at every instant, and compare it with arbitration ablations, a structural variant, a monolithic learned policy, and a planner-style velocity search.

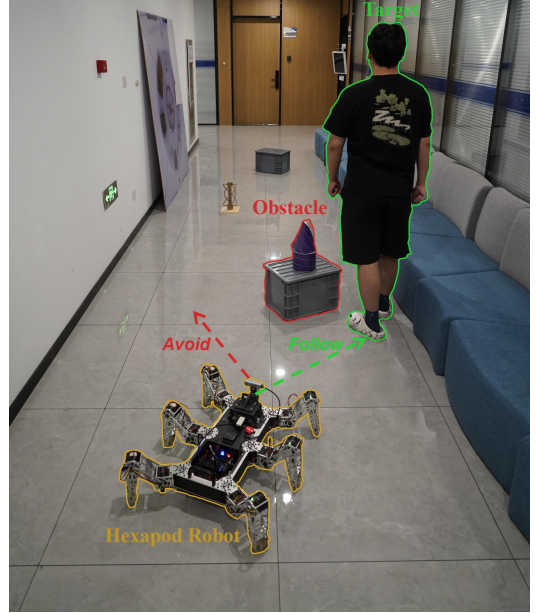


Fig. 1. Real corridor setting for hexapod target following in clutter. The target can pass through narrow gaps that are not safely replicable by the robot’s leg-inclusive safety envelope, creating a command-space conflict between target recovery and collision avoidance.

PCR-Net addresses this conflict directly in the robot’s command space (Fig. 2). It treats the analytic Follow expert and the learned Avoid expert as command proposals in orthogonal subspaces, then uses a learned gate to produce one executable fusion—neither a hard switch nor a predicted detour—conditioned on a per-command risk. The risk is a command-direction-conditioned local clearance prior, not a prediction of the world; it informs the gate rather than deciding for it. All arbitration runs on top of a fixed, pretrained low-level locomotion policy shared by every compared method. The hexapod platform is a useful testbed for this arbitration problem: omnidirectional stepping makes lateral sidestepping natural, while each command must still be realized by a fixed 18-joint locomotion policy rather than a perfect velocity-controlled base.

Although command-space conflict is not specific to row geometries, we use staggered rows as a controlled stress test: alternating narrow gaps repeatedly require lateral avoidance while the target advances, isolating the arbitration problem for systematic ablation.

Contributions. This work makes three contributions.

- **Command-space reframing of target following in clutter.** We formulate following in structured clutter

as a command-level conflict between target-preserving and collision-avoiding motion, and instantiate it through an orthogonal Follow/Avoid command decomposition whose structural properties keep the arbitration bounded and interpretable.

- **Risk-conditioned soft arbitration for hexapod following.** We propose PCR-Net, a high-level module that couples per-command risk with context-dependent fusion of the Follow and Avoid commands while using a fixed low-level locomotion policy.
- **Controlled validation against rule-based, structural, learning, and planner-style alternatives.** Through simulation ablations, an additive-fusion variant, a monolithic policy, and a DWA-style velocity search, together with a quantitative hexapod deployment, we characterize the resulting safety-following trade-offs and show that learned arbitration achieves the most favorable safety-following balance among the tested alternatives in our benchmark.

II. RELATED WORK

Following, avoidance, and agile locomotion. Early follow-and-avoid systems separate the two behaviors with hand-tuned priorities or finite-state logic, and person-following is largely framed as mobile-robot tracking and local navigation [1]. Closest in setting, obstacle-avoidant quadruped leader following has been studied with navigation-centric designs [2]. We instead isolate the continuous command-space arbitration needed by a legged follower under passability mismatch. Recent learned legged controllers achieve agile avoidance and robust perceptive locomotion by coupling control with local navigation [3], [4]: He et al. [3] switch an agile and a recovery policy through a learned reach-avoid value for collision-free high-speed motion toward a *static* goal. We follow a *moving* person through clutter, where the difficulty is preserving tracking during avoidance, and we fuse continuously rather than switching discretely.

Planning, safety, and expert arbitration. Dynamic-window and velocity-space methods [5] score short-horizon rollouts against clearance and progress costs; we adapt a DWA-style target-aware search as a planner-style alternative (Sec. VI), sharing our map, velocity limits, and low-level stack but with a fixed hand-designed cost. Task-priority and null-space control allocate lower-priority objectives to residual freedom [6], while control-barrier-function QPs encode safety constraints in online optimization [7]. Reach-avoid value learning [8] provides a safety-oriented alternative, while mixture-of-experts models provide a classical mechanism for gated expert composition [9], and shared-autonomy methods contextualize policy blending [10]. Residual policy learning augments nominal control with a learned action correction [11]; PCR-Net instead confines the residual to the scalar arbitration coordinate between fixed expert proposals. Here the human is the moving target; PCR-Net uses command-direction-conditioned clearance as a prior for a learned gate, and resolves the conflict by continuous fusion rather than by switching, null-space projection, or trajectory search.

III. PROBLEM FORMULATION

A. Setting. We instantiate the general clutter-following problem in a representative staggered-row layout (Fig. 1), where alternating gaps repeatedly expose the passability mismatch. The high-level controller outputs a velocity command $u = [u_x, u_y, u_\omega]^\top$ (lateral, forward, yaw) at 10 Hz, realized by a fixed low-level locomotion policy at 50 Hz. The robot perceives a local affordance map and the relative target state; it has no global map.

B. Orthogonal command decomposition. We assign following and avoidance to orthogonal command subspaces. The Follow expert acts in the forward-yaw subspace and the Avoid expert in the lateral subspace,

$$\mathcal{U}_F = \{[0, u_y, u_\omega]^\top\}, \quad \mathcal{U}_A = \{[u_x, 0, 0]^\top\}, \quad (1)$$

so that $\mathbb{R}^3 = \mathcal{U}_F \oplus \mathcal{U}_A$ and $\mathcal{U}_F \perp \mathcal{U}_A$. A Follow command $u_F = [0, u_{F,y}, u_{F,\omega}] \in \mathcal{U}_F$ is computed analytically; the learned Avoid expert produces a raw command projected onto the lateral subspace, $u_A = [u_{A,x}, 0, 0] \in \mathcal{U}_A$. Because yaw lies in the Follow subspace, a nonzero following weight preserves part of the target-oriented yaw during lateral avoidance, helping the robot keep facing the target while it sidesteps; whether the target is actually retained is examined empirically in Sec. VI.

C. The conflict. When the target passes through an obstacle configuration the robot cannot replicate, the following-optimal command (along u_F) and the avoidance-optimal command (along u_A) lie in orthogonal subspaces, and a single command generally cannot maximize both target recovery and immediate clearance. The controller must therefore *arbitrate* between them rather than select one.

D. Task success and diagnostics. We separate event success from trajectory diagnostics. A trial succeeds only if, at the success event, the robot has cleared the final obstacle row, lies within the follow band, and has stayed collision-free over the episode. We then report trajectory diagnostics: follow MAE (mean $|\text{dist}(\text{target}, \text{robot}) - d_{\text{des}}|$ over high-level steps), L1 (no collision), L2 (target retained: follow MAE ≤ 0.5 m and in-FOV ≥ 0.90), and L3 (final-row clearance). L1/L2/L3 are not additional success requirements; a successful trial can have MAE above 0.5 m if transient errors occur during obstacle negotiation and the robot recovers before the event (Sec. VI-A).

IV. METHOD

PCR-Net is a three-level hierarchy (Fig. 2): a fixed locomotion policy tracks velocity commands, two experts propose commands in orthogonal subspaces, and a risk-conditioned gate fuses them into one executable command.

A. Follow expert (analytic). The Follow expert computes $u_F = [0, u_{F,y}, u_{F,\omega}]$ analytically from the relative goal, using a turn-in-place-then-advance rule with hysteresis: when the bearing exceeds 0.35 rad the forward term is suppressed and the robot turns in place; it resumes advancing once the bearing falls below 0.18 rad.

B. Avoid expert (learned). The Avoid expert takes the local affordance map, robot state, and relative goal, and

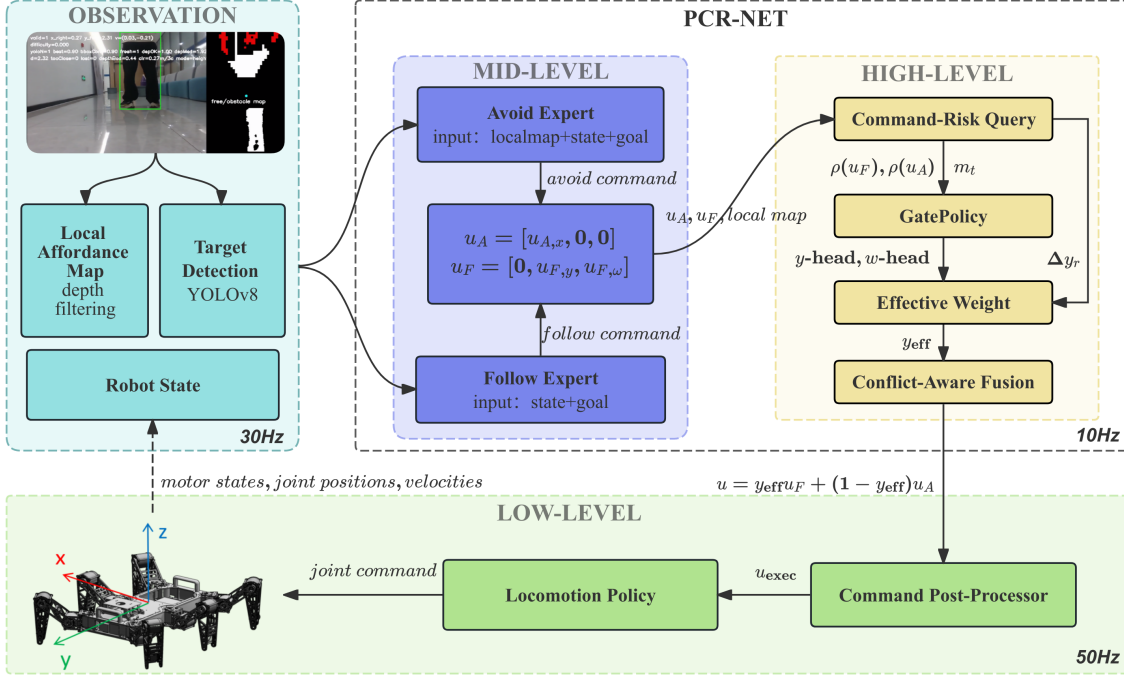


Fig. 2. PCR-Net command-space arbitration. Onboard perception supplies a local occupancy–clearance map and relative target state (YOLOv8 [12]). The analytic Follow expert proposes forward–yaw motion, the learned Avoid expert proposes lateral motion, and command-conditioned risks inform a learned gate that computes y_{eff} and blends the two proposals. A shared fixed locomotion policy tracks the shaped command. Rates indicate deployment frequencies: onboard perception updates at 30 Hz, arbitration runs at 10 Hz, and the fixed locomotion policy tracks commands at 50 Hz.

produces a raw three-dimensional command projected onto the lateral subspace (u_A) before arbitration. It is pretrained in the structured obstacle-row scenes and then frozen (Sec. IV-F), so all arbitration methods share the same avoidance capability.

C. Per-command risk (PCR). For a candidate command u , we evaluate a risk by a cone query on the local affordance map M_t (32×32 cells, 0.09375 m per cell, channels: occupancy and a clearance proxy). Along the command bearing $\text{atan2}(u_x, u_y)$ we open a cone of half-angle 25° and take the minimum clearance d within it ($d = 3.0$ m if no obstacle). The risk is

$$\rho(u) = 1 - \text{clip}\left(\frac{d - d_{\text{safe}}}{d_{\text{free}} - d_{\text{safe}}}, 0, 1\right), \quad (2)$$

with $d_{\text{safe}} = 0.27$ m, $d_{\text{free}} = 0.57$ m; we evaluate the directional gate risks $\rho_F = \rho(u_F)$ and $\rho_A = \rho(u_A)$. This ρ is a *command-direction-conditioned clearance prior*: it depends on the command bearing and the nearest occupied cell in the cone rather than a predicted trajectory. The prior is direction-conditioned rather than speed-aware. As a sensitivity check, we also evaluate $\rho_v(u) = \rho(u) \min(1, v_{\text{cmd}}/v_{\text{max}})$, where $v_{\text{cmd}} = \|[u_x, u_y]\|_2$. This speed-scaled variant produces only marginal differences over 0.35–0.60 m/s, supporting the simpler directional form in the tested low-speed regime.

Distance-decayed risk memory. To retain awareness of a hazard that briefly leaves the field of view, we maintain a risk memory that decays with measured forward body motion rather than elapsed time (it does not decay during purely

lateral motion or while stopped):

$$\begin{aligned} m_t &= \max(\rho_{F,t}, m_{t-1} e^{-\Delta s/L_{\text{clear}}}), \\ \Delta s &= \max(v_{y,t}^{\text{body}}, 0) \Delta t, \end{aligned} \quad (3)$$

with $L_{\text{clear}} = 0.40$ m. The risk memory is a temporal feature supplied to the gate; it does not replace the instantaneous risks ρ_F, ρ_A used by the analytic correction below.

D. Gate and conflict-aware fusion. A single GatePolicy uses Beta heads for the base following weight y and learned conflict variable $w \in (0, 1)$, matching their bounded semantics while allowing state-dependent concentration; the centered transform of w gives the signed modulation Δy_w . It encodes the local affordance map, robot state, relative goal, and the command-conditioned features (the candidate commands u_F, u_A and their difference, the per-command risks ρ_F, ρ_A , clearances, command cosine, conflict score, and risk memory). The symbol y denotes this scalar following gate, whereas u_y denotes the forward component of a velocity command. The effective following weight combines a learned and an analytic correction,

$$\Delta y_w = \lambda \text{deadband}(2w - 1), \quad \Delta y_r = \gamma(\rho_A - \rho_F), \quad (4)$$

$$y_{\text{eff}} = \text{clip}(y + \Delta y_w + \Delta y_r, 0, 1), \quad (5)$$

with $\lambda = 0.30$, $\gamma = 0.15$, and $\text{deadband}(z) = z$ for $|z| > 0.05$ and 0 otherwise. The fused command is the convex blend

$$u = y_{\text{eff}} u_F + (1 - y_{\text{eff}}) u_A. \quad (6)$$

Values $w > 0.5$ reinforce following, whereas $w < 0.5$ suppresses it through the signed correction Δy_w . This correction is monotone non-decreasing in its effect on y_{eff} : $\partial y_{\text{eff}} / \partial w = 2\lambda = 0.60$ in the active unsaturated region, and 0 inside the dead-zone or after clipping. The two corrections play distinct roles: Δy_r is an analytic directional risk-difference term, while Δy_w is a learned signed modulation. The naming follows the gate’s configuration: with only the y -head the gate reduces to the *Y-only* baseline; adding the w -head and the command-conflict features yields the *Learned-w* model, our PCR-Net.

E. Low-level locomotion interface. The high-level command is emitted at 10 Hz and tracked by a fixed locomotion policy at 50 Hz. The low-level policy is shared by all compared methods and is never updated during arbitration training; it executes velocity commands but makes no avoidance or following decision. Before reaching it, every method’s fused command passes through the same fixed shaping stage (amplitude clamping, slew-rate limiting, clearance-based scaling); because this stage and the low-level policy are identical across methods, they do not confound the attribution of performance differences.

F. Training procedure. Training proceeds in stages (Fig. 2, bottom to top). First, we pretrain a velocity-conditioned hexapod locomotion policy with RSL-RL [13], mapping high-level velocity commands to joint actions; it is shaped for camera stability and impact smoothness, and is frozen thereafter. Second, we train the Avoid expert alone in the structured obstacle-row scenes to produce a safe lateral command from the local occupancy–clearance map and the relative goal; it is then frozen. The Follow expert uses no network, generating its candidate command analytically. With the experts fixed, we train only the arbitration gate: at each step the experts generate candidates, and the gate fuses them using the local map, target state, command conflict, and per-command risk. The final Learned- w model additionally learns the command-conditioned conflict prior w that corrects the base following weight. A 0.05-weighted binary-cross-entropy auxiliary term on the w head favors $w = 0$ on high-confidence avoidance states: row-unreleased cases satisfying $\rho_F > 0.25$, $\rho_F - \rho_A > 0.05$, and command cosine < 0.5 , plus a stricter global fallback with higher risk and margin thresholds. Thus, the centered transform of w serves as a signed, context-dependent correction: it can reinforce or compensate the analytic risk term Δy_r rather than acting as a one-directional avoidance trigger; its realized sign is examined in the hardware trace. No expert or low-level policy is jointly fine-tuned, so all arbitration methods share identical expert and locomotion capability.

The gate is trained with PPO [14] in Isaac Gym [15] over a two-dimensional curriculum of target speed and obstacle-row count. Each episode samples one of four levels—two rows at 0.25–0.40 m/s, two rows at 0.35–0.50 m/s, three rows at 0.30–0.50 m/s, and four-to-five rows at 0.35–0.50 m/s—with the target speed held constant within an episode. Over 120,000 episodes the sampling shifts from easier to harder levels while retaining a fraction of easy episodes to reduce

forgetting. Each obstacle stage has complementary left-first and right-first zigzag layouts; at every reset a layout is chosen reproducibly and each obstacle is perturbed by up to 0.06 m along both horizontal axes, rejecting perturbations that violate passage feasibility. Training obstacles are fixed cylinders (0.15 m radius, 0.50 m height); contact friction is randomized in $[0.4, 0.8]$, with no mass, push, gravity, or actuator randomization. The policy input during training is an actor-rasterized occupancy–clearance map restricted to the D435i stereo FOV rather than rendered depth. The gate input is a two-channel occupancy–clearance map plus 13 robot-state scalars, two target-state scalars, and 16 conflict features; encoders feed a 256–256 ELU trunk with Beta heads for y and w . PPO uses 256 envs, 48 steps/iteration, five epochs, minibatch 12288, learning rate 6×10^{-5} , entropy 0.04, clip 0.05, and 1000 iterations. Main reward terms are row progress, follow distance, success, collision/terminal penalties, and the w auxiliary term.

G. Arbitration design rationale. The command risk is used as a clearance prior rather than a reach-avoid value, and unlike CBF-based controllers it does not provide a formal forward-invariance certificate. Instead, three design choices keep the one-step fusion bounded and interpretable. (i) The Follow and Avoid subspaces form an orthogonal direct sum ($\mathbb{R}^3 = \mathcal{U}_F \oplus \mathcal{U}_A$), so following and lateral avoidance are separable and no command axis is lost; for given (u_F, u_A) the output is a one-dimensional convex trade-off, not an arbitrary vector. (ii) For any $y_{\text{eff}} \in [0, 1]$ the fused command (6) lies in the convex hull of $\{u_F, u_A\}$, so PCR-Net never extrapolates beyond the two proposals, unlike a per-axis gate or additive fusion (Sec. VI-C). (iii) The fusion is a residual policy on the convex expert segment. Let $u_{\text{nom}} = yu_F + (1 - y)u_A$ denote the nominal command and $\delta u = u - u_{\text{nom}}$. Then

$$\delta u = (y_{\text{eff}} - y)(u_F - u_A), \quad \|\delta u\| \leq (\lambda + \gamma)\|u_F - u_A\|. \quad (7)$$

Thus, corrections are disagreement-aligned, vanish at coincident proposals, and have bounded authority during conflict. PCR-Net therefore implements bounded residual policy blending over a convex expert segment.

V. EXPERIMENTS

A. Setup. We evaluate on the fixed hardest stage (five rows, 13 capsule slots, minimum passage width 0.75 m, close to the robot’s ~ 0.65 m leg-inclusive safety envelope) and sweep three discrete target speeds $\{0.35, 0.50, 0.60\}$ m/s. The first two lie within the training range; 0.60 m/s is above the maximum training speed of 0.50 m/s, so the high-conflict condition is an above-training speed-extrapolation evaluation. Main arbitration results are mean $\pm$ std over three randomized evaluation runs of 128 episodes each (384 episodes per method–speed setting), with a 50-s episode horizon, on the fixed hardest stage. The inference ablation and DWA-style search use that stage; monolithic PPO is evaluated over increasing row difficulty. The hexapod model has 18 actuated joints, a $0.20 \times 0.44 \times 0.05$ m body collision box, leg-link lengths 0.072/0.13/0.17 m, and a 0.25 m

row-passing body half-length; the D435i stereo camera is mounted at (0, 0.22, 0.08) m in the body frame with 0° pitch, 20° roll, 90° yaw, $87^\circ \times 58^\circ$ depth FOV, and 0.28–3.0 m range. Simulation uses structured arrangements of regular geometric obstacle primitives in Isaac Gym.

B. Compared methods. We organize the comparison into three groups. (1) *Arbitration ablations* share the experts, observations, and execution stack but apply different arbitration rules: *Y-only* ($y_{\text{eff}} = y$, no risk or learned modulation); *Geom-w*, a geometric clearance heuristic $w_{\text{geom}} = e^{-d/\tau}$ using the Follow-command clearance d and $\tau = 0.15$ m, with $y_{\text{eff}} = y(1 - w_{\text{geom}})$; *Risk-only*, the analytic Δy_r term only, retrained from scratch with no learned- w head; *Rule-Override*, a reactive override on $\rho_F - \rho_A$ that forces lateral avoidance by cutting the forward command to as low as 10% while retaining up to 70% of the yaw command, using one fixed parameter set across all speeds; and *Learned-w*, the full PCR-Net. (2) *Structural alternative*. *Additive-Fusion* replaces the gate with direct command addition, $u = u_F + u_A$, keeping the same experts but with no gate. (3) *External alternatives* (Sec. VI): a *monolithic PPO* policy and a *DWA-style velocity search*, both using the same local map, relative-target input, command limits, and fixed low-level stack.

C. Metrics. Task success is the event-level criterion of Sec. III-D; we report collision rate and row-progress as the L1/L3 diagnostics, plus follow MAE and L2 where target retention is relevant.

VI. RESULTS

All simulation results use the event-level success definition of Sec. III-D; their evaluation protocols are specified in Sec. V-A.

A. Main performance. At high conflict the methods separate by their safety–following balance: among the tested arbitration designs, PCR-Net is the only one that pairs high task success with low collision (Table I; Fig. 3). Fig. 4(a) shows the fixed high-conflict scene and perception panels. At 0.35 m/s the methods are less separated; at 0.50 m/s, Y-only has already collapsed while the remaining methods retain success of 0.737–0.992. At the above-training speed 0.60 m/s, Learned-w (ours) is the only method above 0.90 success while keeping collision below 0.05, sustaining task success 0.953, collision 0.039, and follow MAE 0.623 m (the MAE exceeds the 0.5 m L2 threshold without contradicting the high success, per Sec. III-D). Y-only and the geometric heuristic fall to near-zero success; Risk-only drops to 0.008 success, with collision 0.294 and follow MAE 1.305 m, indicating failure to recover following.

At 0.60 m/s, Geom-w provides insufficient avoidance authority (0.003 success, 0.427 collision). Rule-Override reaches 0.734 success but incurs 0.211 collision, over five times Learned-w’s rate; at 0.35 m/s, the same fixed rule over-triggers and reduces success to 0.646, compared with 0.987 for Learned-w. These results show that neither one-sided attenuation nor a fixed override maintains the safety–following balance across speeds, whereas Learned-w achieves the strongest tested combination of high task completion and

TABLE I
MAIN PERFORMANCE ON THE FIXED HARDEST STAGE (MEAN $\pm$ STD).

Spd Method	Succ. $\uparrow$	Row $\uparrow$	Coll. $\downarrow$	MAE [m] $\downarrow$	
0.35	Y-only	0.979 $\pm$ 0.005	0.996 $\pm$ 0.001	0.021 $\pm$ 0.005	0.217 $\pm$ 0.001
	Geom-w	0.969 $\pm$ 0.014	0.992 $\pm$ 0.006	0.031 $\pm$ 0.014	0.171 $\pm$ 0.001
	Risk-only	0.964 $\pm$ 0.012	0.993 $\pm$ 0.002	0.036 $\pm$ 0.012	0.164 $\pm$ 0.001
	Rule-Ov.	0.646 $\pm$ 0.063	0.821 $\pm$ 0.040	0.346 $\pm$ 0.066	0.164 $\pm$ 0.008
	Additive	0.654 $\pm$ 0.012	0.842 $\pm$ 0.010	0.344 $\pm$ 0.014	0.102$\pm$0.001
	Learned-w	0.987$\pm$0.009	0.998$\pm$0.001	0.013$\pm$0.009	0.120 $\pm$ 0.000
0.50	Y-only	0.000 $\pm$ 0.000	0.930 $\pm$ 0.012	0.307 $\pm$ 0.064	0.969 $\pm$ 0.035
	Geom-w	0.935 $\pm$ 0.005	0.972 $\pm$ 0.007	0.065 $\pm$ 0.005	0.430 $\pm$ 0.001
	Risk-only	0.971 $\pm$ 0.009	0.990 $\pm$ 0.005	0.029 $\pm$ 0.009	0.405 $\pm$ 0.001
	Rule-Ov.	0.737 $\pm$ 0.071	0.900 $\pm$ 0.028	0.247 $\pm$ 0.073	0.381 $\pm$ 0.013
	Additive	0.870 $\pm$ 0.009	0.946 $\pm$ 0.004	0.125 $\pm$ 0.008	0.177$\pm$0.002
	Learned-w	0.992$\pm$0.008	0.999$\pm$0.002	0.008$\pm$0.008	0.355 $\pm$ 0.003
0.60	Y-only	0.000 $\pm$ 0.000	0.794 $\pm$ 0.009	0.714 $\pm$ 0.036	1.133 $\pm$ 0.012
	Geom-w	0.003 $\pm$ 0.005	0.746 $\pm$ 0.018	0.427 $\pm$ 0.084	1.172 $\pm$ 0.015
	Risk-only	0.008 $\pm$ 0.008	0.715 $\pm$ 0.002	0.294 $\pm$ 0.071	1.305 $\pm$ 0.031
	Rule-Ov.	0.734 $\pm$ 0.055	0.861 $\pm$ 0.027	0.211 $\pm$ 0.048	0.582 $\pm$ 0.019
	Additive	0.669 $\pm$ 0.123	0.874 $\pm$ 0.050	0.326 $\pm$ 0.126	0.334$\pm$0.040
	Learned-w	0.953$\pm$0.028	0.985$\pm$0.008	0.039$\pm$0.027	0.623 $\pm$ 0.004

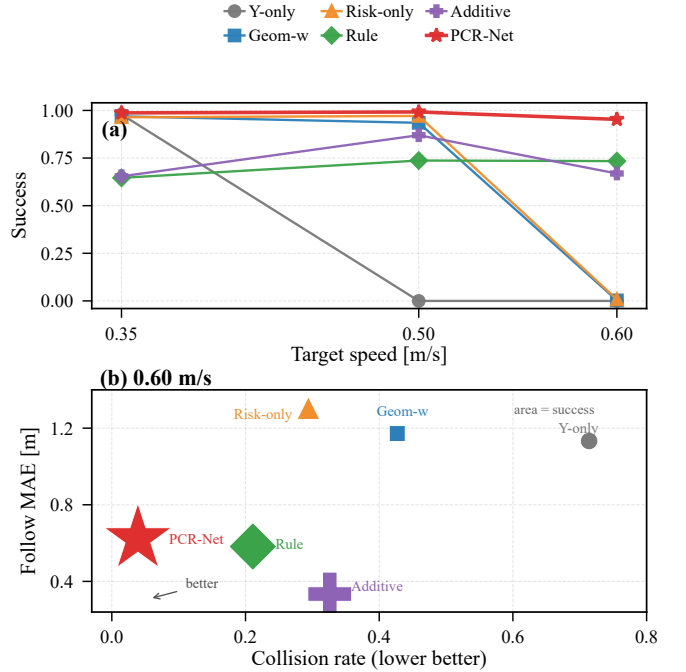
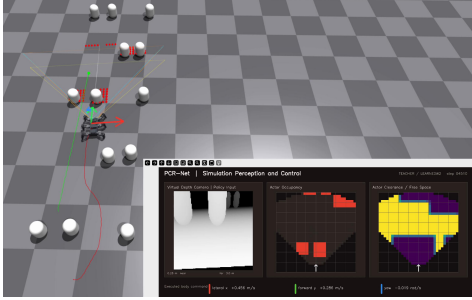


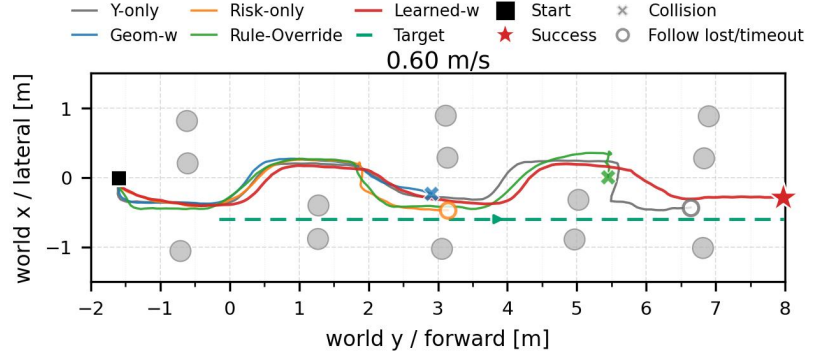
Fig. 3. Main performance at the fixed hardest stage. (a) Task success vs. target speed. (b) Collision rate vs. follow MAE at 0.60 m/s (point area encodes task success; lower-left is better).

low collision. Y-only’s nontrivial row-progress at 0.60 m/s should not be read as successful navigation, since it comes with a 0.714 collision rate.

B. Inference ablation. Table II tests the learned correction by reusing the same Learned-w checkpoint but forcing Δy_w to zero only at inference, leaving all else unchanged. Within the training range (0.35 and 0.50 m/s), removing Δy_w retains high task success; the collapse appears only at the above-training speed 0.60 m/s. At this speed, removing Δy_w reduces row progress from 0.985 to 0.739 and raises follow MAE from 0.623 to 1.463 m, while collision is 0.133. The ablated policy still advances but cannot sustain the coordi-



(a) Simulation scene and perception inputs.



(b) Representative trajectories at 0.60 m/s.

Fig. 4. Simulation evidence at the hardest stage (0.60 m/s). (a) The obstacle-row scene and simulated perception inputs, including occupancy and clearance maps. (b) Geom-w collides at the third row, Risk-only loses the target near the third row, Rule-Override collides at the fourth row, and Y-only reaches the final row but loses the target; Learned-w alone clears all five rows and succeeds.

TABLE II

INFERENCE ABLATION OF THE LEARNED CORRECTION: THE SAME LEARNED-W CHECKPOINT IS REUSED WITH Δy_w FORCED TO ZERO AT INFERENCE. Δy_w IS REPORTED AS PREDICTED/USED AVERAGES.

Spd	Success $\uparrow$	Row-prog. $\uparrow$	Coll. $\downarrow$	MAE [m] $\downarrow$	Δy_w (pred./used)
0.35	0.974	0.993	0.023	0.176	0.066/0
0.50	0.982	0.992	0.016	0.438	0.067/0
0.60	0.000	0.739	0.133	1.463	0.066/0

nated progress needed to maintain follow distance through the final row; its dominant failure is therefore loss of follow-distance regulation rather than collision. This checkpoint-controlled ablation measures the trained composite policy's dependence on Δy_w and indicates that the learned correction supplies the high-conflict margin needed for sustained following at the above-training boundary.

C. Structural alternative: Additive-Fusion. Does the orthogonal decomposition alone, without a learned gate, already resolve the conflict? Additive-Fusion ($u = u_F + u_A$) preserves tracking well (lowest follow MAE at 0.35/0.50 m/s) but collides far more (0.344, 0.125, 0.326) and is unstable at high speed (0.669 success at 0.60 m/s): preserving following is easy, but staying safe needs context-dependent arbitration of how much avoidance to inject, which direct addition lacks. Fig. 4(b) shows representative high-conflict trajectories.

D. External alternatives. We include two external alternatives with matched base observations, command space, velocity limits, command post-processing, and fixed low-level policy. Published leader-following and agile-navigation stacks often couple different sensors, planners, and low-level controllers. We therefore use interface-matched alternatives to isolate the arbitration design. Monolithic PPO also matches PCR-Net in network capacity and training budget; the DWA-style search uses validation-only tuned costs. They test whether a single learned policy or hand-designed local search can replace learned command-space arbitration in this benchmark.

TABLE III

MONOLITHIC PPO AT FIXED TARGET SPEED 0.35 m/s OVER INCREASING OBSTACLE-ROW DIFFICULTY.

Stage	Success $\uparrow$	Row-prog. $\uparrow$	Coll. $\downarrow$	FOV-in	L2	TTC [s]
2	0.093	0.424	0.445	0.244	0.180	6.49
3	0.064	0.322	0.688	0.313	0.375	4.21
4	0.000	0.016	1.000	0.922	0.898	1.51

Monolithic PPO. As a structural alternative, we train a monolithic PPO policy with the same base observations, matched backbone capacity, and the same training budget as PCR-Net, but without the Follow/Avoid decomposition, the directional risk query, or the gate; it predicts the high-level command directly (Table III, evaluated at 0.35 m/s over increasing row difficulty). It records only 0.093/0.064 success at Stages 2/3 and falls to 0.000 at Stage 4, where collision reaches 1.000; the high Stage-4 FOV-in/L2 reflect short pre-collision episodes (time-to-contact 1.51 s), so the failure is early collision rather than target loss. Under matched capacity and budget, the unstructured policy fails to form a viable safety-progress baseline as row difficulty increases.

DWA-style velocity search. For the planner-style external baseline, we implement a target-aware DWA-style velocity-space search under the same local-map observation, relative-target state, high-level command space, and shared post-processor as PCR-Net. It samples commands $u = [u_x, u_y, u_w]$, applies a dynamic-window filter from the previous command using acceleration limits, rolls out each reachable candidate on the local map, rejects candidates whose footprint intersects occupied cells or whose clearance falls below a margin, and scores the rest by risk, target distance, bearing, progress, and command smoothness,

$$J = \alpha_r J_{\text{risk}} + \alpha_d J_{\text{dist}} + \alpha_b J_{\text{bearing}} - \alpha_p J_{\text{prog}} - \alpha_f J_{\text{fwd}} + \alpha_s J_{\Delta u} + \alpha_l J_{\text{lat}}. \quad (8)$$

After validation-only tuning, all tests use a $7 \times 5 \times 5$ base grid augmented with fixed near-zero velocity samples before dynamic-window filtering; rollouts use a 1.2 s horizon at

TABLE IV
DWA-STYLE VELOCITY-SEARCH ALTERNATIVE.

Search bias	Spd	Success $\uparrow$	Coll. $\downarrow$	Row-prog. $\uparrow$	MAE [m] $\downarrow$
Safety-biased	0.35	0.053	0.612	0.138	0.678
	0.60	0.000	0.838	0.000	1.028
Balanced	0.35	0.026	0.938	0.188	0.375
	0.60	0.000	0.969	0.375	0.592
Tracking-biased	0.35	0.082	0.744	0.344	0.528
	0.60	0.000	0.881	0.500	0.690

0.15 s, 0.32 m footprint, 0.08 m clearance margin, and acceleration limits (3.0, 2.0, 4.0) for forward, lateral, and yaw commands. A finer $21 \times 11 \times 11$ resolution check gives the same qualitative safety–progress trend at $4 \times$ mean planning latency; we report the base grid to satisfy the shared 10 Hz high-level control budget. Unlike PCR-Net, this fixed-cost short-horizon search has neither learned context-dependent arbitration nor distance-decayed risk memory; Table IV tests its sufficiency in this matched benchmark, not the broader DWA family. The resulting comparison is reported in Table IV. In this moving-target row-conflict setting, no fixed-cost setting yields a stable safety–progress balance: success is at most 0.082 at 0.35 m/s and zero at 0.60 m/s, with collision 0.612–0.969. The footprint check is applied to nominal rollouts on the local grid, while contact is measured after execution by the shared legged stack; short-horizon feasible commands can therefore still lead to contact after replanning and tracking dynamics. Safety bias suppresses progress yet remains collision-prone; tracking-oriented costs increase progress without resolving collision.

E. Real-robot deployment. We deploy the policy on a hexapod in a cluttered corridor: a narrow indoor corridor with two staggered obstacle rows whose passable space alternates left and right, so the robot must sidestep into the open gap of each row while keeping the walking person in view. Hardware rows replace the regular simulated primitives with boxes and cones, providing direct evidence of transfer beyond the simulated obstacle representation. The robot closes the loop with onboard perception: YOLOv8 target detection [12] provides the relative goal; target-associated depth points are masked out before the remaining stereo depth is rasterized into the two-channel occupancy–free-space map. Under the same staggered-row protocol, PCR-Net succeeds in 19/20 trials versus 12/20 for Rule-Override, whose eight failures comprise six collisions and two target-loss events (Table V). On these matched trials, PCR-Net reduces mean follow-distance MAE from 1.21 to 0.52 m and raises mean executed forward speed from 0.32 to 0.51 m/s, indicating tighter tracking with more sustained progress. The hardware deployment uses stereo-depth maps rather than the actor-rasterized maps used in simulation, and the 60-trial hardware evaluation tests this deployed perception path directly; to reduce transient perception loss the deployment adds a short obstacle-map memory to the scalar risk memory of Sec. IV-C.

Fig. 5 shows a representative logged real-robot segment.

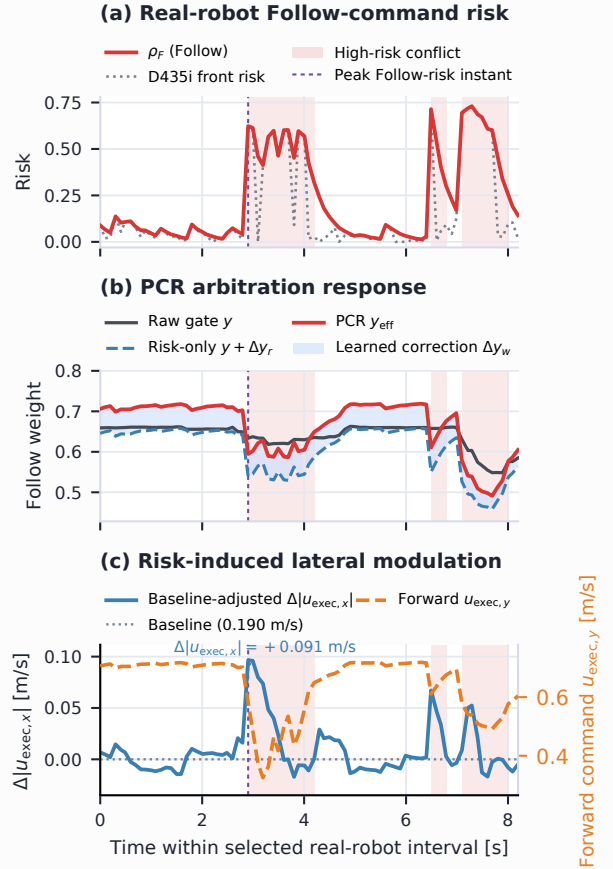


Fig. 5. Representative real-robot arbitration trace. (a) Stereo-derived risk peaks mark conflict intervals. (b) PCR lowers the risk-only Follow response and the learned term partially restores y_{eff} . (c) The executed command increases lateral motion while keeping forward motion nonzero.

Stereo-derived Follow-risk peaks lower the risk-only Follow weight, and the learned correction partially restores y_{eff} while keeping it below the raw gate, so the net command remains avoidance-biased without fully suppressing Follow. During these conflict intervals, the lateral command rises while forward motion stays nonzero. The hardware trace complements the high-conflict simulation in Fig. 4(b): risk-conditioned arbitration keeps forward motion active during lateral avoidance rather than collapsing into a pure avoidance override. Fig. 6 shows the deployed robot sequence and the corresponding onboard perception/local-map sequence.

We additionally evaluate a more demanding corridor scene that requires a sharp target turn between obstacle rows: the target exits the first row, turns 90° , and enters a second row with a different traversal direction. The same deployed policy succeeds in 18/20 trials without scene-specific switching, with failures from one glancing obstacle contact and one target-loss event during the turn. The complete sharp-turn sequence is included in the supplementary video. Across both scenes PCR-Net succeeds in 37/40 trials, providing quantitative hardware evidence that the deployed perception–arbitration–locomotion chain can operate on the real hexapod under corridor clutter. Each hardware trial is scored at

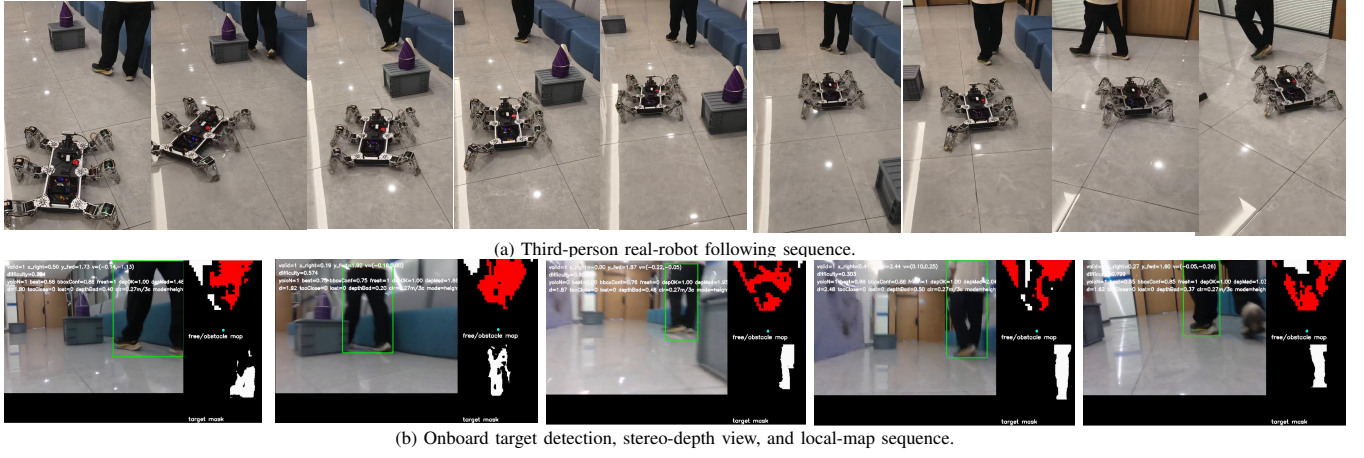


Fig. 6. Real-robot deployment views. (a) The hexapod follows the walking target while sidestepping around staggered corridor obstacles. (b) The onboard perception stream provides the target state and stereo-depth local affordance map used by PCR-Net.

TABLE V
REAL-ROBOT RESULTS OVER 60 TRIALS. RULE-OVERRIDE USES
STAGGERED ROWS; THE SHARP-TURN SCENE JOINS TWO ROW
SEGMENTS AT 90°.

Method	Scene	Trials	Success	Coll.	Lost	Rate
PCR-Net	Staggered rows	20	19	1	0	95%
Rule-Ov.	Staggered rows	20	12	6	2	60%
PCR-Net	Sharp target turn	20	18	1	1	90%
PCR-Net	Overall	40	37	2	1	92.5%

the episode level: success requires completing the corridor segment without obstacle contact or target loss.

VII. DISCUSSION AND LIMITATIONS

Why it works. PCR-Net retains following authority during lateral avoidance. Removing Δy_w reduces row progress and breaks follow-distance regulation rather than primarily causing collision. The alternatives expose complementary failures: additive fusion cannot regulate avoidance authority, while fixed-cost velocity search finds no stable safety-progress balance in this benchmark.

Limitations and future work. Controlled simulation ablations use straight staggered rows, while the sharp-turn hardware scene evaluates two row segments joined by a 90° change in traversal direction, using the same deployed local-map policy without scene-specific switching. The speed-scaled risk check supports the simpler directional prior over the tested low-speed regime; braking-distance- or trajectory-aware risk remains a natural extension for faster motion.

VIII. CONCLUSION

We formulated hexapod target following in clutter as command-space conflict arbitration and proposed PCR-Net, a risk-conditioned gate that softly fuses Follow/Avoid proposals over a fixed locomotion policy. Learned arbitration gives the best tested safety-following balance, with inference ablation identifying the high-conflict role of Δy_w . A 60-trial hardware evaluation (40 PCR-Net, 20 Rule-Override)

across straight and 90° connected-row layouts supports the deployed perception-arbitration-locomotion chain.

REFERENCES

- [1] M. J. Islam, J. Hong, and J. Sattar, “Person-following by autonomous robots: A categorical overview,” *The International Journal of Robotics Research*, vol. 38, no. 14, pp. 1581–1618, 2019.
- [2] C. Scheidemann *et al.*, “Obstacle-avoidant leader following with a quadruped robot,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025, pp. 1407–1413.
- [3] T. He, C. Zhang, W. Xiao, G. He, C. Liu, and G. Shi, “Agile but safe: Learning collision-free high-speed legged locomotion,” in *Robotics: Science and Systems (RSS)*, 2024.
- [4] T. Miki, J. Lee, J. Hwangbo, L. Wellhausen, V. Koltun, and M. Hutter, “Learning robust perceptive locomotion for quadrupedal robots in the wild,” *Science Robotics*, vol. 7, no. 62, p. eabk2822, 2022.
- [5] D. Fox, W. Burgard, and S. Thrun, “The dynamic window approach to collision avoidance,” *IEEE Robotics & Automation Magazine*, vol. 4, no. 1, pp. 23–33, 1997.
- [6] Y. Nakamura, H. Hanafusa, and T. Yoshikawa, “Task-priority based redundancy control of robot manipulators,” *The International Journal of Robotics Research*, vol. 6, no. 2, pp. 3–15, 1987.
- [7] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, “Control barrier function based quadratic programs for safety critical systems,” *IEEE Transactions on Automatic Control*, vol. 62, no. 8, pp. 3861–3876, 2017.
- [8] K.-C. Hsu, V. Rubies-Royo, C. J. Tomlin, and J. F. Fisac, “Safety and liveness guarantees through reach-avoid reinforcement learning,” in *Robotics: Science and Systems (RSS)*, 2021.
- [9] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, “Adaptive mixtures of local experts,” *Neural Computation*, vol. 3, no. 1, pp. 79–87, 1991.
- [10] A. D. Dragan and S. S. Srinivasa, “A policy-blending formalism for shared control,” *The International Journal of Robotics Research*, vol. 32, no. 7, pp. 790–805, 2013.
- [11] T. Johannink *et al.*, “Residual reinforcement learning for robot control,” in *2019 IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 6023–6029.
- [12] G. Jocher, A. Chaurasia, and J. Qiu, “Ultralytics YOLOv8,” Ultralytics, 2023. [Online]. Available: <https://github.com/ultralytics/ultralytics>
- [13] N. Rudin, D. Hoeller, P. Reist, and M. Hutter, “Learning to walk in minutes using massively parallel deep reinforcement learning,” in *Proceedings of the 5th Conference on Robot Learning (CoRL)*, 2022, pp. 91–100.
- [14] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [15] V. Makovychuk *et al.*, “Isaac Gym: High-performance GPU-based physics simulation for robot learning,” in *NeurIPS Datasets and Benchmarks Track*, 2021.